

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_060850_o
- started_at_local: 2026-05-20 06:08:50
- finished_at_local: 2026-05-20 06:42:59
- duration_hhmm: 00:34
- duration_seconds: 2049.033
- seed_offset: 14000
- replicate_label: o
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.091021	0.0	0.057922	0.859479	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235667	0.054968	0.121542	0.81601	0.445	False	0.00304	0.208165
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.046321	0.115856	0.857489	0.445	False	0.009939	0.707203
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.054968	0.121317	0.832569	0.4425	False	0.001236	-0.013835

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.86711	0.4375	False	0.007161	1.104421
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.0	0.4375	False	0.009806	0.48603

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.091021, family holdout 0.0, combined 0.057922.
- Accepted novelty 0.859479, complexity 0.76, and adaptation efficiency 0.053925.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.81601, complexity 0.445, and adaptation efficiency 0.009963.
- Last-epoch runtime 45.883375 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.00304, positive scaffolds 1, Pareto runtime inflation 0.208165.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.173945, "gap_delta": 0.000224, "runtime_inflation": 0.208165, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.857489, complexity 0.445, and adaptation efficiency 0.008531.
- Last-epoch runtime 59.863475 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.009939, positive scaffolds 1, Pareto runtime inflation 0.707203.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.784605, "gap_delta": -0.005462, "runtime_inflation": 0.707203, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.832569, complexity 0.4425, and adaptation efficiency 0.004765.
- Last-epoch runtime 67.611125 ms and distance evaluations 5165.75.
- Validation: surviving False, transplant delta 0.001236, positive scaffolds 0, Pareto runtime inflation -0.013835.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 11, "complexity_score": 0.42, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.013835, "same_gap_faster": true}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.86711, complexity 0.4375, and adaptation efficiency 0.016741.
- Last-epoch runtime 43.6948 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.007161, positive scaffolds 1, Pareto runtime inflation 1.104421.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.865064, "gap_delta": -0.003304, "runtime_inflation": 1.104421, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.0, complexity 0.4375, and adaptation efficiency 0.0.
- Last-epoch runtime 65.07795 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.009806, positive scaffolds 1, Pareto runtime inflation 0.48603.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.357171, "gap_delta": 0.000224, "runtime_inflation": 0.48603, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite Review

Key Evaluation Criteria Summary

Criterion	Priority	Evidence Source(s)
Held-out TSPLIB gap	High	final_tsplib_gap from modular conditions

Criterion	Priority	Evidence Source(s)
Family holdout gap	High	final_family_gap and family_signal
Transplant validation	High	transplant_mean_gap_delta, positive_scaffold_count, transplant_results
Ablation validation	High	ablation_results, effect of disabling operator
Pareto tradeoffs	Medium-High	pareto_gap_delta, pareto_runtime_inflation
Surviving candidates status	Critical	surviving_candidate flag
Reusable operator structure	Preferred	Operator reproducibility and validation across scaffolds/families

Overall Findings

Baselines and Full Solver

- ****Baseline heuristic (phase7_baseline_heuristic_only)****: Final TSPLIB gap is large (0.235058), family gap 0.054968.
- ****Full solver evolution (phase7_full_solver_evolution)**** improves transfer (0.0579) and TSPLIB gap (0.0910) strictly better than baseline.
- However, full solver conditions do not involve modular operators, yielding no reusable operator candidates.

Modular Operators Overview

Operator Name	Type	Final TSPLIB Gap	Family Gap	Transfer Gap	Transplant Mean Gap Delta	Ablation Signal	Surviving Candidate	Comments
xfer_candidate_prune_by_trap_cluster_v1	candidate_pruner	0.235667	0.054968	0.121542	0.00304	**No effect**	False	No improvement over host; disables harmless; high runtime inflation (~0.21)
det_restart_ctrl_twocluster_bottleneck_boost	restart_controller	0.235058	0.046321	0.115856	0.009939	**No effect**	False	Gains on clustered/two-cluster families; no TSPLIB improvement; high runtime (~0.71)
restart_controller_two_cluster_safe	restart_controller	0.235058	0.050827	0.118702	0.007161	**No effect**	False	Helps "two_cluster_bottleneck_tsp" family slightly; no TSPLIB impact; highest runtime inflation (~1.1)
acceptance_bounded_worse_with_stagnation_safe_guard	acceptance	0.235058	0.054968	0.121317	0.001236	**No effect**	False	No family or TSPLIB improvements; disables harmless; runtime slightly lower (-0.013)

Operator Name	Type	Final TSPLIB Gap	Family Gap	Transfer Gap	Transplant Mean Gap Delta	Ablation Signal	Surviving Candidate	Comments
pruner_cluster_bottleneck_adaptive	candidate_pruner	0.235667	0.054968	0.121542	0.009806	**No effect**	False	Similar to other pruners; no family/TSPLIB gains; disables harmless; moderate runtime inflation (~0.49)

Key Observations

1. **Held-out TSPLIB Gap & Family Transfer**

- None of the modular operators improve TSPLIB gaps relative to the baseline (about 0.235).
- Family holdout gaps remain similar or worse; no clear family benefit except small gains on two-cluster and clustered family subsets by restart controllers.

2. **Transplant Validation**

- All operators show small positive average transplant gap deltas (~0.001 to 0.01), indicating no consistent transferable improvement.
- Positive scaffold counts are low (max 1), and some scaffolds show regressions.
- Operators fail to transplant cleanly into some scaffolds (especially clustered_local_search and sparse_three_opt).

3. **Ablation Validation**

- Disabling operators generally does **not degrade** performance.
- Variant and disabled ablation gaps are nearly identical or even slightly better without the operator.
- This indicates the discovered operators do not produce independent, measurable positive effect.

4. **Pareto Tradeoffs**

- No operator moves clearly on the Pareto frontier improving gap at same runtime.
- Runtime inflation is significant (up to 1.1x) with minimal or no gap improvement.
- Some operators show slight negative gap deltas but at large runtime cost.

5. **Validation of Reusability and Survival**

- None of the operators survive the transplant, ablation, scaffold, or family transfer tests.
- No operators flagged as surviving_candidate.
- Operators do not show reusable modular structure validated by ablation or transplant.

Conservative Conclusions

- ****No discovered modular operator passed key validations to be considered reusable or independently beneficial.****

- The slight transfer or family gains on two-cluster or clustered TSP instances by restart-controller operators exist but do not generalize or improve held-out TSPLIB or survive ablation/transplant checks.
- Candidate pruners similarly fail ablation and transplant tests, indicating lack of real effect.
- High runtime inflation undermines practical utility of these operators.
- Full solver condition improves performance but does not yield modular reusable operators.
- **Hence, no modular-operator discovery or reusable operator structure can be credibly claimed from these results.**
- Narrative speculation unsupported; focus should remain on improving transfer and ablation validation criteria.

Recommendation

- Continue refining modular discovery methods focusing on:
- Ensuring ablation disables operator cause detectable performance drop.
- Improving transfer to held-out families and TSPLIB datasets.
- Achieving operator survival after transplant and family holdout.
- Reducing runtime inflation to Pareto-beneficial levels.
- Given current results, do not promote any operator as reusable or as a discovered algorithm component.

Summary

Status	Result
Modular operators found	Yes, several candidates
Survive transplant test	No
Survive ablation test	No
Improve held-out TSPLIB	No
Improve family holdout	Marginal/sparse, not robust
Pareto runtime/gap trade	Poor (runtime inflation high)
Survive family holdout	No clear survivors
Declared discovered algos	None
Recommended action	Further refinement needed